The short answer is: to try and become a better software developer. Knowing about data structures and algorithms is like adding a tool to your tool belt.

Have you ever heard of Maslow’s hammer, aka the law of the instrument? It’s a conjecture, driven by observation, about how people who only know one way to do things, tend to apply what they know to all kind of different situations.

If your tool belt only has a hammer, you will be tempted to treat everything as a nail. If you only know how to sort a list, you will be tempted to sort your tasks list every time you add a new task or have to pick the next one to tackle, and you will never be able to leverage the context to find more efficient solutions.

The purpose of this book is giving you many tools you can use when approaching a problem. We will build upon the basic algorithms normally presented in a computer science 101 course (or alike) and introduce the reader to more advanced material.

After reading this book you should be able to recognize situations where you could improve the performance of your code by using a specific data structure and/or algorithm.

Obviously, your goal should not be remembering by heart all the details of all the data structures we will discuss. We will rather try to show you how to reason about issues, where to find ideas about algorithms that might help you in solving problems. This book will also serve as a handbook, sort of a recipe collection, with indications about some common